Physical Features of India

The big idea

India has every kind of landform: the lofty **Himalayas**, the flat fertile **Northern Plains**, the ancient **Peninsular Plateau**, the sandy **Indian Desert**, long **Coastal Plains** and two groups of **islands**. Each was formed differently and each supports a different way of life.

How the landforms were formed

According to the **Theory of Plate Tectonics**, the Earth's crust is made of large plates that move. Millions of years ago the **Indo-Australian plate** broke off from the super-continent **Gondwana land** and drifted north, colliding with the much larger **Eurasian plate**. The sedimentary rocks of the **Tethys** sea between them were folded up to form the **Himalayas**. The deep depression left was filled by sediments from rivers, forming the **Northern Plains**. The **Peninsular Plateau** is the oldest landmass, a part of Gondwana land, made of igneous and metamorphic rocks.

TYPES OF PLATE BOUNDARIES

Convergent (plates come together; fold mountains like the Himalayas), **Divergent** (plates move apart), **Transform** (plates slide past each other).

The Himalayan mountains

The Himalayas are **young fold mountains** stretching about **2,400 km** in an arc from the Indus to the Brahmaputra. Their width is **400 km in Kashmir** to **150 km in Arunachal Pradesh**. They have three parallel ranges:

1. **Great or Inner Himalayas (Himadri):** the most continuous range, average height **6,000 m**, with the loftiest peaks; core of granite; perennially snow-bound; many glaciers. Highest peaks: Mt. Everest (8,848 m), Kanchenjunga (8,598 m), Nanga Parbat, Nanda Devi.
2. **Lesser Himalaya (Himachal):** altitude **3,700–4,500 m**, average width 50 km. Ranges: **Pir Panjal** (longest), **Dhauladhar** and **Mahabharat**. Famous valleys: **Kashmir, Kangra and Kullu**, and well-known hill stations.
3. **Shiwaliks (Outer Himalayas):** altitude **900–1,100 m**, width 10–50 km; made of unconsolidated sediments. Longitudinal valleys between the Himachal and the Shiwaliks are called **Duns** (Dehra Dun, Kotli Dun, Patli Dun).

Regional divisions (from west to east, by river valleys):

- **Punjab Himalaya**: between Indus and Satluj (also Kashmir and Himachal Himalaya); Karakoram, Ladakh, Zaskar, Pir Panjal ranges.
- **Kumaon Himalaya**: between Satluj and Kali.
- **Nepal Himalaya**: between Kali and Teesta.
- **Assam Himalaya**: between Teesta and Dihang.

East of the **Dihang gorge** (Brahmaputra), the Himalayas bend south and spread along India's eastern boundary as the **Purvachal** or Eastern hills: **Patkai hills, Naga hills, Manipur hills and Mizo hills**, mostly sandstone.

The Northern Plains

Formed by three river systems, the **Indus, Ganga and Brahmaputra** and their tributaries. It covers about **7 lakh sq km**, about 2,400 km long and 240–320 km wide. It is densely populated and agriculturally very productive.

- **Punjab Plains** (west): formed by the Indus and its tributaries; much of it lies in Pakistan. Has many **doabs** (doab = land between two rivers; "do" = two, "ab" = water).
- **Ganga Plains**: between the Ghaggar and Teesta rivers; spread over north Indian states, Haryana, Delhi, UP, Bihar, part of Jharkhand and West Bengal.

- **Brahmaputra Plains**: in Assam.

Relief features (north to south):

Belt	Feature
Bhabar	Narrow belt, 8–16 km wide, at the foot of the Shiwaliks; pebbles deposited; streams disappear
Terai	Streams re-emerge; wet, swampy, marshy; thick forests and wildlife (much cleared for agriculture)
Bhangar	Older alluvium, above flood level; contains calcareous deposits called **kankar**
Khadar	Newer, younger alluvium of flood plains; renewed every year; very fertile

Riverine islands (like **Majuli** in the Brahmaputra, the largest inhabited riverine island) and **distributaries** are also features.

The Peninsular Plateau

A **tableland** of old crystalline, igneous and metamorphic rocks; formed from Gondwana land. It has two divisions:

- **Central Highlands**: north of the **Narmada**; bounded by the **Vindhyan range** in the south and the **Aravalis** in the north-west. Wider in the west, narrower in the east. Eastward extensions are the **Bundelkhand and Baghelkhand**; the **Chotanagpur plateau** is drained by the Damodar. Rivers here flow from south-west to north-east.
- **Deccan Plateau**: triangular, south of the Narmada; bounded by the **Satpura range** in the north, the **Mahadev, Kaimur hills and Maikal range** in the east. It is higher in the west and slopes gently east. An extension is the **Meghalaya (Shillong) plateau** and **Karbi Anglong plateau** in the north-east, separated by a fault from Chotanagpur; Garo, Khasi and Jaintia hills.

Western Ghats vs Eastern Ghats:

Western Ghats	Eastern Ghats
Lie parallel to the western coast	Stretch from Mahanadi valley to the Nilgiris in the south
Continuous; can be crossed only through passes (**Thal, Bhor and Pal Ghats**)	**Discontinuous** and irregular; dissected by rivers draining into the Bay of Bengal
Higher: average 900–1,600 m	Lower: average 600 m
Highest peaks: **Anai Mudi (2,695 m)**, **Doda Betta (2,637 m)**	Highest peak: **Mahendragiri (1,501 m)**
Cause orographic rain from moist winds	Receive less rain

The **Eastern and Western Ghats meet at the Nilgiri hills**. The Deccan trap is a region of **black soil** of volcanic origin.

The Indian Desert

Lies to the west of the **Aravali hills**. An undulating sandy plain covered with **sand dunes**; low rainfall, **below 150 mm** per year; arid climate with little vegetation. Streams appear in the rainy season and then disappear into the sand. **Luni** is the only large river. **Barchans** (crescent-shaped dunes) cover larger areas; longitudinal dunes are more prominent near the Indo-Pakistan border.

The coastal plains and islands

- **Western coast** (between the Western Ghats and the Arabian Sea): narrow. Three sections: **Konkan** (Mumbai–Goa), **Kannad plain** (central), **Malabar coast** (south).
- **Eastern coast** (along the Bay of Bengal): wide and level. **Northern Circar** (northern part) and **Coromandel Coast** (southern part). Large rivers

(Mahanadi, Godavari, Krishna, Kaveri) form extensive **deltas**. **Chilika Lake** (Odisha) is the largest saltwater lake in India.

- **Lakshadweep Islands**: in the Arabian Sea, close to the Malabar coast, made of small **coral** islands (earlier Laccadive, Minicoy and Amindive); area 32 sq km; administrative headquarters **Kavaratti**. Pitti island is uninhabited and has a bird sanctuary.
- **Andaman and Nicobar Islands**: an elongated chain in the Bay of Bengal, bigger and more numerous; believed to be elevated portions of submarine mountains. Divided into **Andaman** in the north and **Nicobar** in the south. **Barren Island**, India's only active volcano, is here. Rich in biodiversity; equatorial climate.

BHABAR VS BHANGAR

Bhabar is the pebble belt at the foot of the Shiwaliks. **Bhangar** is the older alluvial terrace of the plains. They sound alike but are very different.

MAP POINTS

Mountain ranges (Karakoram, Zaskar, Shiwalik, Aravali, Vindhya, Satpura, Western and Eastern Ghats); peaks (K2, Kanchenjunga, Anai Mudi); plateaus (Deccan, Chotanagpur, Malwa); coastal plains (Konkan, Malabar, Coromandel, Northern Circar).

Quick recap

- Himalayas: young fold mountains with Himadri, Himachal and Shiwalik ranges; Duns; Purvachal in the east.
- Northern Plains: alluvial; bhabar, terai, bhangar, khadar; doabs.
- Peninsular Plateau: oldest; Central Highlands and Deccan; Western Ghats continuous, Eastern Ghats discontinuous.
- Indian Desert: west of Aravalis, barchans, Luni river.

- Coasts: Konkan, Kannad, Malabar (west); Northern Circar, Coromandel (east). Islands: Lakshadweep (coral), Andaman and Nicobar (volcanic origin, Barren Island).

Practice questions

1. The northernmost range of the Himalayas is called
   (a) Himadri (b) Himachal (c) Shiwaliks (d) Purvachal
2. Longitudinal valleys between the Lesser Himalaya and the Shiwaliks are known as
   (a) Doabs (b) Bugyals (c) Duns (d) Khadars
3. The newer, younger alluvium of the flood plains is called
   (a) bhabar (b) bhangar (c) kankar (d) khadar
4. The highest peak in the Eastern Ghats is
   (a) Anai Mudi (b) Doda Betta (c) Mahendragiri (d) Kanchenjunga
5. The Eastern and Western Ghats meet at the
   (a) Aravali hills (b) Nilgiri hills (c) Garo hills (d) Cardamom hills
6. The only active volcano of India is located on
   (a) Lakshadweep (b) Majuli (c) Minicoy (d) Barren Island
7. The southern part of the east coastal plain is called
   (a) Konkan (b) Malabar (c) Coromandel Coast (d) Northern Circar
8. Distinguish between the Western Ghats and the Eastern Ghats. Also compare the Himalayan region with the Peninsular Plateau.
9. Write a short note on the Indian Desert.

Answers: 1. a 2. c 3. d 4. c 5. b 6. d 7. c 8. see notes 9. see notes